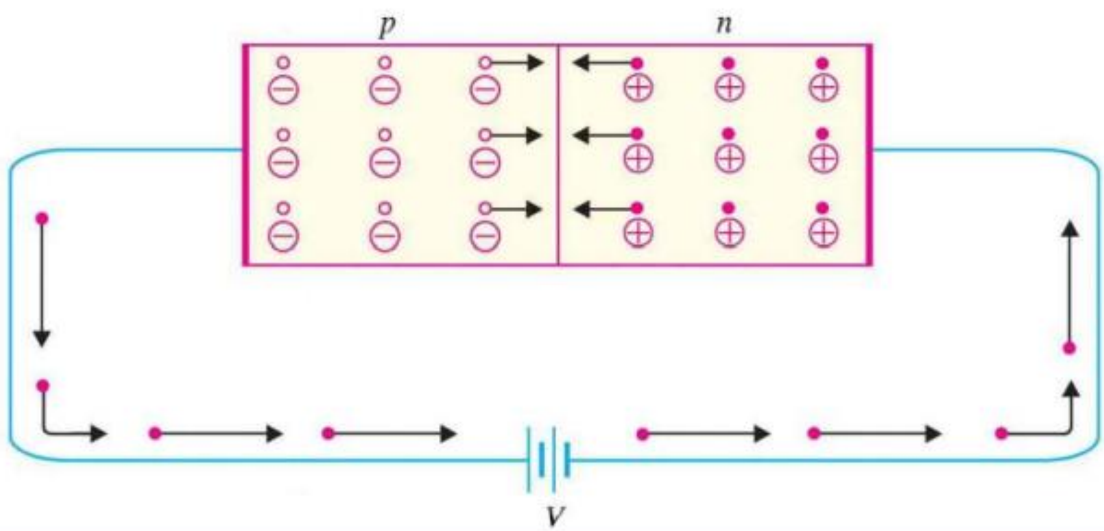
1st lecture (13/03/2024)

- **Suggested book:** Modern Digital Electronics- R.P. Jain (2nd edition)
- **Diode:**
  - **What and How its work?**
    A diode is a semiconductor device that essentially acts as a one-way switch for current.
    A diode is like a one-way street for electricity. It lets electricity flow in one direction but stops it from going the other way. It works by using two different types of materials that create a barrier, allowing current to pass through when electricity is applied in one direction, but blocking it when applied in the opposite direction.
- **Transistor.**
  - **Forward bias:** Forward bias or biasing is where the external voltage is delivered across the P-N junction diode. In a forward bias setup, the P-side of the diode is attached to the positive terminal, and N-side is fixed to the negative side of the battery.



  - **Reverse Bias:** Reverse bias is when the p-side of the diode is connected to the negative voltage of the battery and the n-side is connected to the positive voltage of the battery. Figure opposite connection as forward bias**.**
  - **Why it is called transistor?**
    The term "transistor" is a combination of "transfer" and "resistor." A transistor is called so because it transfers electrical signals across a resistor.

✪ **Active, Cut-off and Saturation region of Transistor.**

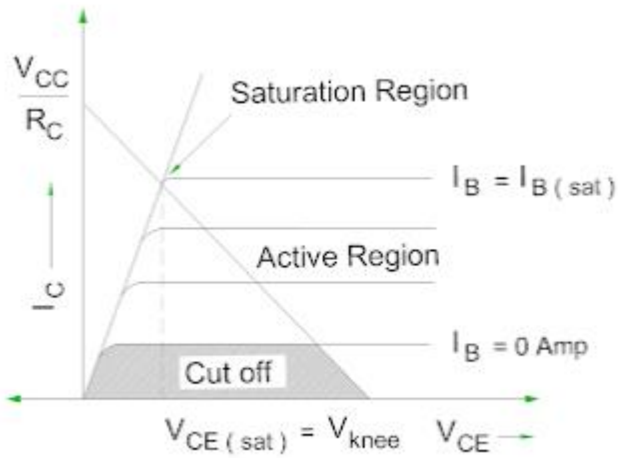


FIG : CUT OFF, ACTIVE AND SATURATION REGION OF TRANSISTOR

✪ [**Transistor gates: AND, OR, NAND, NOR.**]

2nd lecture (18/03/2024)

✪ **Resistor–transistor logic (RTL),** sometimes also known as transistor–resistor logic, is a class of digital circuits built using resistors as the input network and bipolar junction transistors as switching devices.

✪ The basic RTL gate is a NOR gate as shown in Fig. 4.4. For the sake of simplicity, a two-input NOR gate driving N similar gates shown in the figure, which can be extended to accommodate a larger number of inputs. The number of input terminals is referred to as the fan-in.

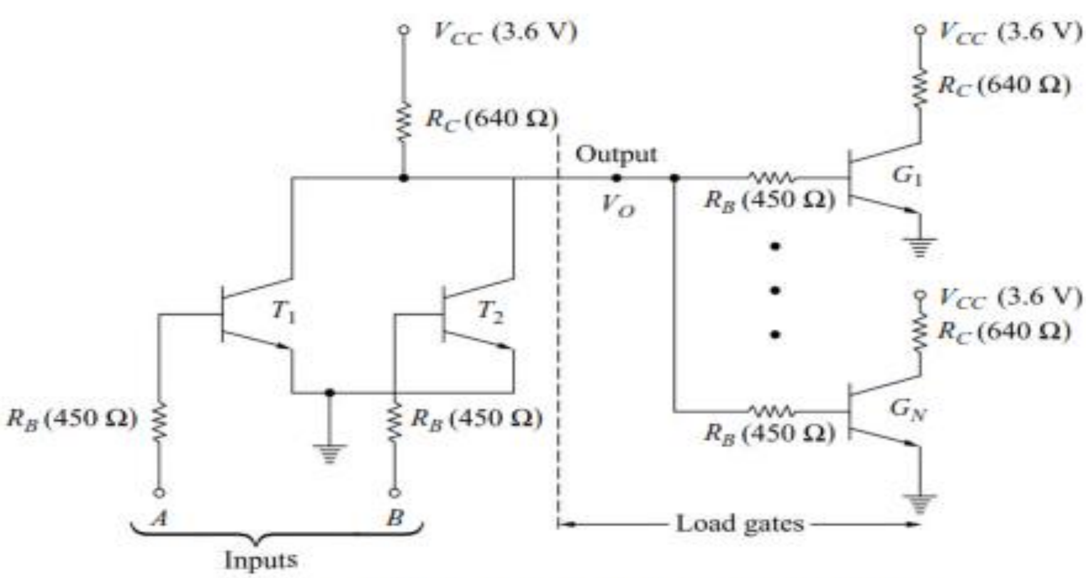


Fig. 4.4 *A 2-input RTL* NOR *Gate Driving N Similar Gates*

If all the inputs to the gate are LOW, the output is HIGH and if the gate is not driving any other gate, i.e. no load is connected, the output voltage will be slightly less than V_{CC} (there is voltage drop across the common collector resistor due to I_{CO} of T_1 and T_2).

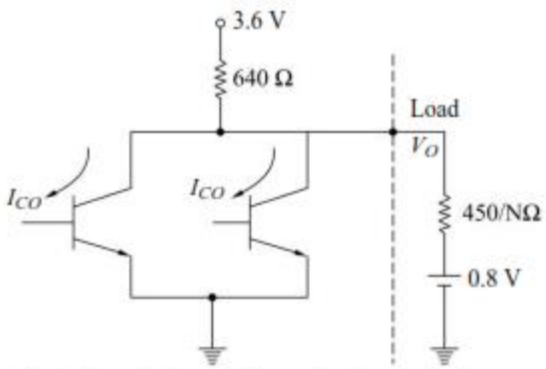


Fig. 4.5 ***A Circuit Illustrating the Equivalent Circuit at the Input of the Load Gates***

When N similar gates are being driven, the load will be equivalent to a resistor of value 450/N ohms in series with a voltage source of 0.8 V (being the voltage between base and emitter of a transistor in saturation). The relevant portion of the circuit is shown in Fig. 4.5.

The base current for each load transistor is

$$I_B = \left(\frac{3.6 - 0.8}{640 + \frac{450}{N}} \right) \cdot \frac{1}{N} = \frac{2.8}{640N + 450} \quad (4.1)$$

The collector current for the load transistor in saturation is

$$I_{C,\text{sat}} = \frac{3.6 - 0.2}{640} = 5.31 \text{ mA} \quad (4.2)$$

The value of N must satisfy the following relation,

$$h_{FE} \cdot I_B \geq I_{C,\text{sat}} \quad (4.3)$$

For $N = 5$, $I_B = 0.767$ mA. Therefore, h_{FE} must be greater than 7.

Noise margin.

When the output is in 0 state V= 0.2 V. If this voltage becomes about 0.5 V (cut-in voltage of transistor), the load transistor comes to conduction which causes malfunction of the circuit. Hence, the logic 0 noise margin ∇0=0.3V.

The logic 1 noise margin depends upon the number of gates being driven. For $N = 5$,

$$V_o = \frac{90}{90 + 640} \times (3.6) + \frac{640}{90 + 640} \times (0.8) = 1.14 \text{ V} \quad (4.4)$$

For $h_{FE} = 10$, the total base current required for load transistors to be driven into saturation will be $5 \times \left(\frac{5.31}{10} \right)$ mA and the corresponding V_o must be 1.04 V. Therefore, the noise margin for 1 level is $\Delta 1 = 1.14 - 1.04 = 0.1$ V.

- ✪ [Click here and look over two input RTL NAND gate.]
- ✪ [Click here and look over two input RTL NOR gate.]

3rd lecture (20/03/2024)

- **Diode-transistor logic (DTL)** belongs to the digital logic family. This logic circuit has diodes at the input side and transistor at the output side and so the name diode transistor logic.
- ***Introduction to DTL: Click here and look over the video***
- ***DTL NAND gate: Click here and look over the video.***
- ***DTL NOR gate: Click here and look over the video*.**
- For the DTL NAND gate of Fig. 4.12 calculate (a) fan-out (b) noise-margins, and (c) average power, P, dissipated by the gate. The diode and transistor parameters are:

  Diode:

  Voltage across a conducting diode = 0.7 V

  Cut-in voltage V = 0.6 V

  Transistor:

  Cut-in voltage V = 0.5 V

  $V_{BE,sat}$=0.8V

  $V_{CE,sat}$=0.2V

  h_{PE}=30

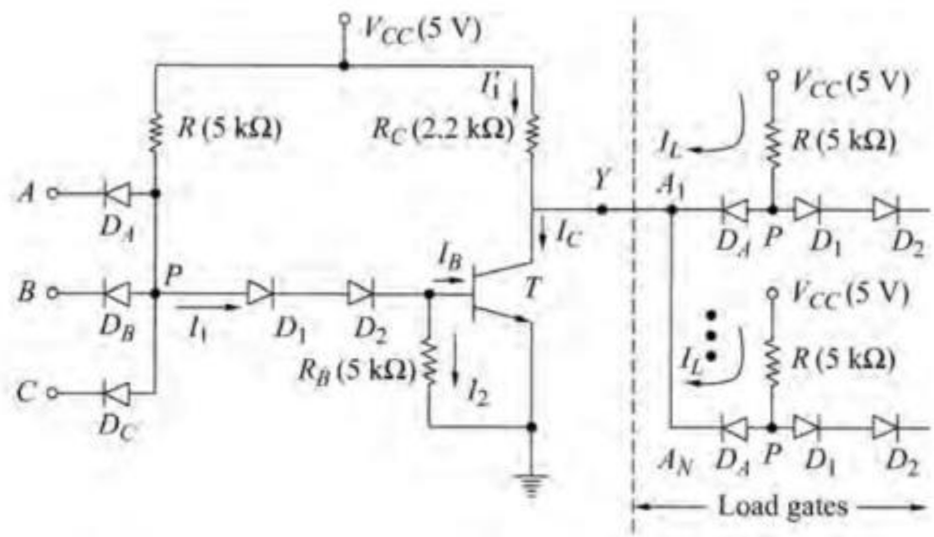


Fig. 4.12 ***A 3-Input DTL* NAND *Gate Driving* N *Similar Gates***

Solution

(a) As discussed above, the logic levels are:

$$\text{LOW level} = V(0) = V_{CE,sat} = 0.2\text{ V}$$
$$\text{HIGH level} = V(1) = V_{CC} = 5\text{ V}$$

(i) If all the inputs are HIGH, the input diodes are reverse-biased. Assuming diodes D_1, D_2 to be conducting and T to be in saturation, the voltage $V_p = 0.7 + 0.7 + 0.8 = 2.2$ V.

Writing Kirchhoff's current law (KCL) equation at the base of T,

$$I_B = I_1 - I_2$$

where

$$I_1 = \frac{V_{CC} - V_P}{R} = \frac{5 - 2.2}{5} = 0.56 \text{ mA}$$

and

$$I_2 = \frac{V_{BE,\text{sat}}}{R_B} = \frac{0.8}{5} = 0.16 \text{ mA}$$

which gives a base current $I_B = 0.4$ mA. The collector current (without load gates connected) is

$$I_C = \frac{V_{CC} - V_{CE,\text{sat}}}{R_C} = \frac{5 - 0.2}{2.2} = 2.182 \text{ mA}$$

Since $h_{FE} \times I_B = 30 \times 0.4 = 12$ mA is greater than I_C (2.182 mA), it is confirmed that the transistor is in saturation and the output is in LOW state. Now, if N load gates are fed from this gate, the input diodes of the driven gates will conduct through the output transistor T, i.e. T acts as a sink for the current in the input to the gates it drives. Assuming that all the other inputs to each of the load gates are HIGH except the one driven by T, the current $I_L = \frac{V_{CC} - V_P}{R} = \frac{5 - 0.9}{5} = 0.82$ mA. This current is referred to as *standard load*. The fan-out is given by $I_C \le h_{FE} I_B$, or $0.82\,N + 2.182 \le 12$ mA or $N < 12$ since N must be an integer. A conservative choice is $N = 10$. The Maximum collector current rating of T must be about 12 mA.

(ii) If at least one of the inputs is LOW, the corresponding input diode conducts and $V_P = 0.2 + 0.7 = 0.9$ V. The minimum voltage required for D_1, D_2, and T to be conducting is $0.6 + 0.6 + 0.5 = 1.7$ V, which confirms that D_1, D_2 are nonconducting and hence T is cut-off. Consequently, the output voltage is V_{CC} (5 V) if the load gates are not connected.

If the load gates are connected, the input diodes of the load gates are nonconducting, which means the reverse-saturation current of these diodes must be supplied through the collector resistor R_C, which will produce a voltage drop across R_C and consequently the output voltage corresponding to HIGH state will be a little less than V_{CC}. The maximum current which can be supplied by the gate will depend upon V_{OH}. The fan-out is determined on the basis of maximum current.

(b) (i) If all the inputs are HIGH, the output is LOW. Since $V_P = 2.2$ V, the input diodes are reverse-biased by $5 - 2.2 = 2.8$ V. Since the cut-in voltage of the diode is 0.6 V, a negative noise spike of at least 3.4 V present at the input will cause malfunction of the circuit, i.e. the 0 level noise margin $\Delta\, 0 = 3.4$ V.

(ii) If at least one input is LOW, the output is HIGH. Since $V_P = 0.9$ V and a voltage of at least 1.7 V [part a (*ii*)] is required for D_1, D_2, and T to conduct, therefore a positive noise spike of at least 0.8 V will cause malfunction of the circuit, i.e. the 1 level noise margin $\Delta\, 1 = 0.8$ V.

(c) The power $P(0)$ when the output is LOW is given by $P(0) = V_{CC}(I_1 + I_1') = 5(0.56 + 2.182) = 13.71\,\text{mW}$. When the output is in the HIGH state at least one of the input diodes conduct. Therefore, $I_1 = 0.82$ mA and $I_1' = 0$ Hence $P(1) = (0.82)(5) = 4.1$ mW.

If we assume that the occurrence of LOW and HIGH is equally likely then the average power is

$$P_{\text{av}} = \frac{P(0) + P(1)}{2} = \frac{13.71 + 4.1}{2} = 8.905\,\text{mW}$$

4th lecture (23/04/24)

- **Transistor–Transistor logic (TTL)** is a logic family built from bipolar junction transistors.
- A 3-Input TTL NAND Gate : **Click here and look over the video.**
- A TTL Gate with Totem-Pole Output Driver: **Click here and look over the video.**

5th lecture (28/04/24)

- **Clippers.**
  - Series and shunt circuits.
- Click here and look over the video from 37 to 40.
- Solve related example from book: Clippers start from page 78.

6th lecture (05/05/24)

- **Clampers.**
  - Positive and negative.
- Click here and look over the video
- Solve related example and exercise from book.

7th lecture (13/05/24)

- Operational Amplifier (OP-Amp).
- Circuit symbols and terminal of OP-Amp 741C.
- Block diagram for OP-Amp in sensor base data acquisition.
- Inverting OP-Amp configuration. (09 page of pdf)

8th lecture (15/05/24)

- Derive equation from figure 3.1
- Related example-3.1 and 3.2.
- Inverting adder, pdf page-23.